## AI3403: Multi-Agent Systems - Group Project Proposal

# Resilient 3D Aerial Traffic Management and Knowledge-Gated Coordination for UAV Fleets in Disaster Response

**Group Name: Aerial**

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## 1. Problem and Motivation

In a disaster zone a fleet of UAVs must sweep damaged sectors and find survivors with the ground communication infrastructure down and no central controller. Each UAV talks only to neighbours within its short wireless range, so the communication graph $\mathcal{G}(t)$ keeps changing as UAVs move and packets are lost. Two things go wrong: links drop at random, and a few UAVs may be compromised and broadcast wrong positions, which breaks ordinary consensus completely. The standard defence is $W$-MSR filtering, where each UAV discards the $F$ largest and $F$ smallest received values before averaging. This restores convergence, but it also throws information away.

That trade-off is the heart of our project. After filtering, a UAV's estimate may say the airspace ahead is clear while several states consistent with what it received are *not* clear, so acting on the estimate alone is unsafe. We argue a UAV should advance only when it can establish the move is safe from what it genuinely observed. We implement this as an *epistemic knowledge gate* above the resilient consensus layer, grade the behaviour with Signal Temporal Logic (STL) robustness, and measure what the caution costs in mission time. In short:

> **The filtering that makes a fleet resilient also makes it ignorant, and we intend to quantify that trade-off.**

## 2. Technical Approach

Three layers sit over a common simulator, each taken directly from the course.

**(a) Coordination.** $N$ UAVs with double-integrator dynamics

$$\dot{p}_i = v_i, \qquad \dot{v}_i = u_i, \qquad p_i \in \mathbb{R}^3.$$

Neighbours come from a distance-bounded disk graph with Bernoulli packet drops, so $\mathcal{G}(t)$ is time-varying. On this we implement the consensus behaviours from the lectures: rendezvous, velocity alignment, formation with offsets, leader–follower pinning and BOIDS flocking.

**(b) Resilience.** Under an $F$-local attack model each agent applies $W$-MSR over its trusted set $\mathcal{R}_i[k]$, giving

$$v_i[k{+}1] = v_i[k] + \alpha \sum_{j \in \mathcal{R}_i[k]} w_{ij}[k]\big(v_j[k] - v_i[k]\big).$$

We also build an $(r, s)$-robustness checker and test the graph conditions on Erdős–Rényi graphs.

**(c) Knowledge.** From the messages that survived and were rejected by filtering, agent $i$ builds a finite set of possible worlds $I_i(w)$: which neighbours could be faulty, plus a coarse interval abstraction of their states. Knowledge is set containment,

$$K_i(\varphi) \equiv I_i(w) \subseteq [\![\varphi]\!],$$

evaluated under $S5$, and the control input is gated on it: a UAV advances only when *every* consistent world is safe, and otherwise holds and re-queries. This is what makes the epistemic layer change behaviour rather than only report on it.

An STL monitor grades each run. Requirements such as

$$\mathbf{G}_{[0,T]}\left(\|p_i - p_j\| \geq d_{\text{safe}}\right) \qquad \text{and} \qquad \mathbf{F}_{[0,T_1]}\left(p_i \in \mathcal{S}_{\text{target}}\right)$$

are scored by the robustness $\rho$, giving one number per experiment to compare across configurations.

## 3. Scope: What We Commit to in Two Months

We have kept the promised scope deliberately modest and bounded. Anything beyond this is Future Scope.

- A Python simulator (`NumPy`, `NetworkX`, `Matplotlib`) for fleets of **up to** $N = 20$ **agents**, in 2D first and then 3D, with time-varying disk graphs and packet drops. Fixed-step integration; no aerodynamics or physical drone model.
- The coordination behaviours above, with a study of how convergence speed varies with the algebraic connectivity $\lambda_2(L)$ across a few standard topologies.
- $W$-MSR filtering with $F$-local attacks limited to $F \leq 2$, covering three attack types: constant offset, slow drift and random spoofing. An $(r, s)$-robustness checker by enumeration, which is acceptable at $N \leq 20$.
- An **offline** STL robustness monitor written from the quantitative semantics taught in class. We are not attempting online control synthesis from STL.
- The epistemic gate with a deliberately **coarse, finite abstraction**: worlds are limited to at most $F$ faulty neighbours together with values on a coarse grid. We will state the abstraction and report its size.
- A comparison of a naive fleet against a knowledge-gated fleet over repeated randomised runs, producing the trade-off curve between caution and mission completion time.
- A Streamlit dashboard, and a final report with reproducible scripts hosted on GitHub.

Across five members this is roughly 60 hours each. We have sized it so every item above has a working version by Week 6, leaving Weeks 7–8 for experiments, writing and slack.

## 4. Individual Contributions

The work splits into five roughly equal parts. Each member owns the implementation, experiments and report section for their part, and the parts are arranged so nobody blocks anybody.

- **Rithvik Jami - Simulator core and graph layer.** UAV kinematics and integration loop, disk-graph construction with packet drops, Laplacian and spectral utilities, the experiment runner and configuration system, repository setup, and final integration of all modules.
- **Praneeth Chamarthy - Coordination algorithms.** Rendezvous, velocity alignment, formation with offsets, leader–follower pinning and BOIDS flocking; step-size and stability checks; the convergence-versus-connectivity study; the controller that switches between behaviours.
- **Gona Sanjana - Resilience and adversaries.** The attack models and attack library, the $W$-MSR filter wired into every coordination primitive, the $(r, s)$-robustness checker, experiments on Erdős–Rényi graphs, and the study of how performance degrades as $F$ grows.
- **Muvvala Sai Ram Suhas - Epistemic layer.** Construction of the possible-worlds set from filtered and

rejected messages, the $S5$ knowledge evaluator including nested knowledge, and wiring the knowledge gate into the controller.

- **Jakkala Naga Rohith Kumar -Specification, analysis and dashboard.** The STL robustness monitor, the mission specifications, the Streamlit dashboard, and the batch experiments and plots including the final trade-off analysis.
- **Prabhu Teja - Mission planning, scenario generation and evaluation.** Design disaster-response scenarios, mission objectives, target/obstacle generation, and initial/target UAV positions, with experiment and random-seed management. Validate baseline scenarios and aggregate mission metrics such as completion time, safety violations, hesitation time, and target coverage.

## 5. How We Will Demonstrate It

The demonstration is a single Streamlit page built around one central image: **two fleets running side by side under an identical attack**, on the left a naive fleet acting on filtered estimates, on the right a knowledge-gated fleet. The naive fleet moves faster at first and then violates safety on spoofed data; the gated fleet hesitates, re-queries, finishes later and stays safe. That contrast is the point of the project and is visible within seconds.

### Live Dashboard

Around it are four live plots:

- consensus and formation error over time;
- STL robustness $\rho(t)$ for both fleets with the zero line marked so violations are obvious;
- a heatmap of which messages $W$-MSR is rejecting; and
- a knowledge matrix showing $K_i(\text{Safe})$ filling in as information spreads.

Sliders control $N$, $F$, packet-drop probability, attack type and gating. A batch mode generates the report figures, chiefly the **caution-versus-performance curve**: one point per configuration, with the naive and gated families as clearly separated groups.

## 6. Timeline

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| **Weeks 1–2** | Simulator, graph layer, first consensus runs. |
| **Weeks 3–4** | Coordination behaviours complete, STL monitor working offline. |
| **Week 5** | Attacks and $W$-MSR integrated, robustness analysis. |
| **Week 6** | Epistemic gate wired into the controller. |
| **Week 7** | Dashboard, experiment sweeps, figures. |
| **Week 8** | Report, presentation and buffer. The buffer is intentional and we will not plan work into it. |